Dhyey Patel

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EDUCATION

Eastern Michigan University *(M.S. in Computer Science | GPA: 4.0/4.0)* Expected Apr 2027
University of Cincinnati *(B.S. in Computer Science, Minor in Mathematics | GPA: 3.744/4.0)* Aug 2020 – May 2025

EXPERIENCE

Eastern Michigan University – Ypsilanti, MI
Web Developer Sep 2025 – Present

- Developing the public-facing scholarship directory for **650+ awards** (pre-launch), integrating EMU's existing IC Command scholarship API with data normalization, cached retrieval, eligibility filtering, and reusable components.
- Migrated **60+ active legacy forms** to the newer Modern Campus Form Module and standardized Request More Information workflows across **500+ academic program pages and 6 colleges**.
- Built a reusable **Registrar calendar component** backed by EMU Today data, replacing manually maintained semester lists and fixing multi-day event expiration and duplicate API results.
- Engineered and launched **3 accessible transfer-planning tools** by transforming articulation-guide data into structured JSON and reusable program-discovery workflows for partner colleges.
- Reworked catalog-link infrastructure across **400+ pages**, reducing annual maintenance from approximately **2 weeks of team effort to ~1 day for 2 people**.

Givaudan – Cincinnati, OH

Frontend Developer Intern *(Co-op)* Apr 2023 – Aug 2023

- Rebuilt **3 ATOM pages in React + MUI**, cutting page development from ~2 weeks to ~3 days; converted an R/MATLAB algorithm to JavaScript, reducing runtime from ~5s to <0.5s while preserving output parity.
- Diagnosed and fixed production data-retrieval and table issues in Insight, restoring a workflow used by internal teams globally.

Web Developer Intern *(Co-op)* Aug 2022 – Dec 2022

- Built internal experiment-creation and response-capture tools using HTML, CSS, and JavaScript, including a reusable numpad, dynamic image/GIF forms, and Base64 media storage.
- Built a Flask admin dashboard for submitted test data with **<1s** retrieval, and D3.js result visualizations that cut load time from **up to 1 minute to <3 seconds**.

RESEARCH AND PUBLICATIONS

Transfer Learning for Avian Bioacoustics under Sparse Positive Labels – *IEEE MLSP 2026*
Dhyey Patel, Yunting Yin | **First Author** | arXiv:2608.03977

- Built target-only baselines and developed a reliability-aware multi-source transfer-learning framework for sparse positive labels, achieving 0.584 macro-AP and 0.860 macro-AUC.

Prompt Injection Detection for Email Agents Through Attack Chain Modeling – *IEEE ICTAI 2026*
Ahmad Hashmi, Dhyey Patel, Yunting Yin | arXiv:2609.30657

- Contributed to attack-chain labeling, experimental pipelines, and robustness evaluation across 461,640 LLMail rows.
- Framework reached 0.406 mean F1 on 5 benchmarks vs. 0.216 for the best off-the-shelf detector under strict thresholds.

PROJECTS

Job Fair Match (Live) | *Next.js, TypeScript, Prisma, PostgreSQL, Neon, Vercel* Sep 2025 – Dec 2025

- Primary developer on a 6-person class team; built authentication with email verification, candidate and employer profiles, and swipe, match, and chat flows.
- Designed a normalized PostgreSQL schema and backend APIs for profiles, jobs, swipes, and matches, using unique constraints to prevent duplicate swipes and matches.

SegLungAI (GitHub) / *TensorFlow/Keras, scikit-learn, NumPy, U-Net, ResNet-50* Aug 2024 – Apr 2025

- Built an end-to-end **neonatal lung MRI segmentation pipeline** covering preprocessing, training, evaluation, and inference workflows.
- Reached **~0.90 mean Dice** and **~0.83–0.85 IoU** on validation data, and packaged reproducible training scripts for repeatable experimentation.

SKILLS

Languages: Python, JavaScript/TypeScript, PHP, SQL
Technologies: React, Next.js, Node.js, Express, Flask, PostgreSQL, PyTorch, TensorFlow/Keras, Git, REST APIs